

Lamiaa's meeting

Mathieu Ouillon

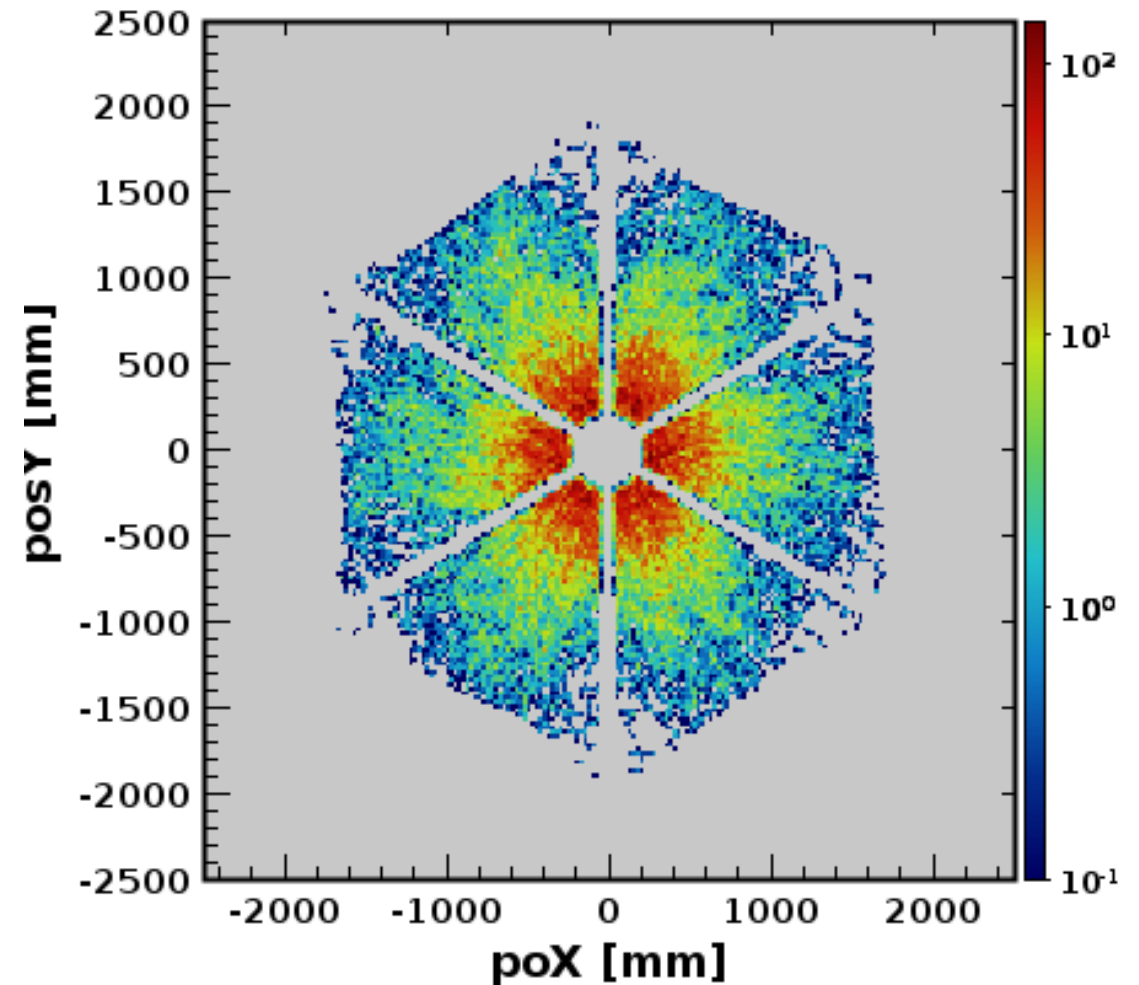
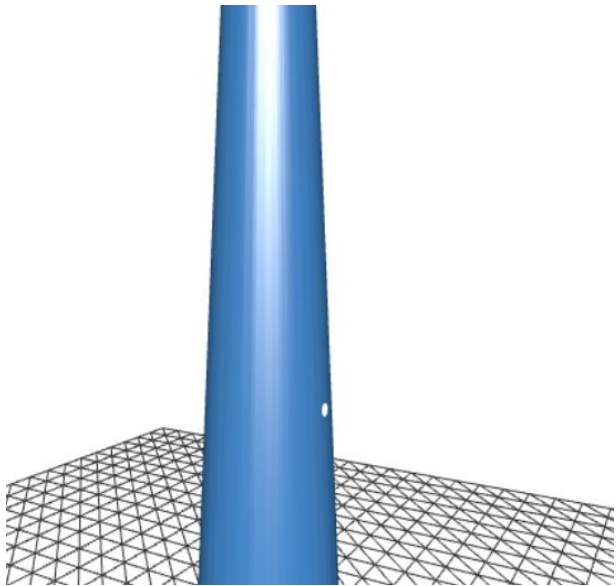
08/27/24



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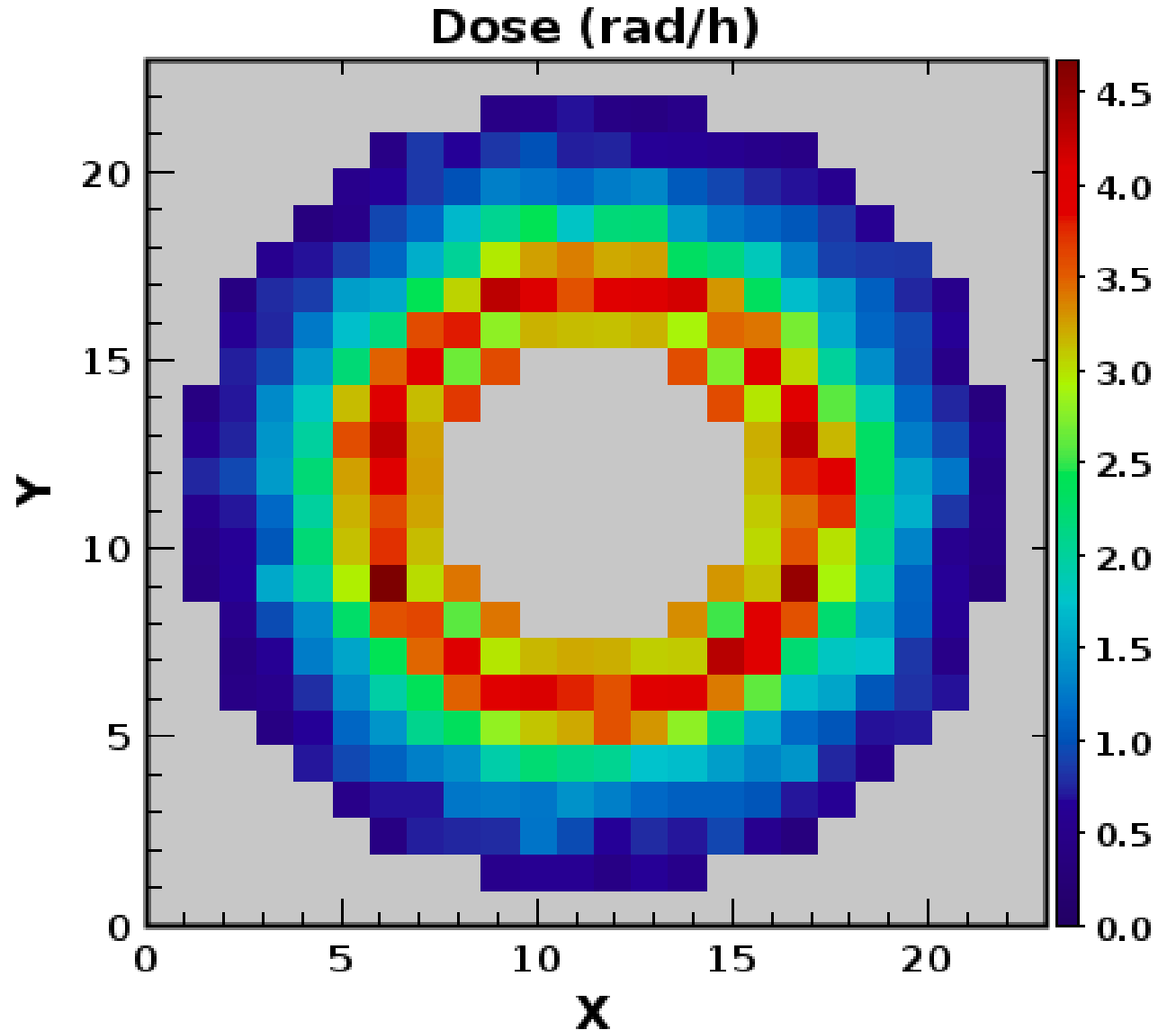
FT and DC occupancy for ALERT

- I have fix the asymmetry in the position of the background in the first region of the DC.
- For that I have change the moller cone for it's cad model to the old one, which is just geant4 volume.
- In the old one there are no hole for the screw, which are the cause of the asymmetry.



FT dose for ALERT setup

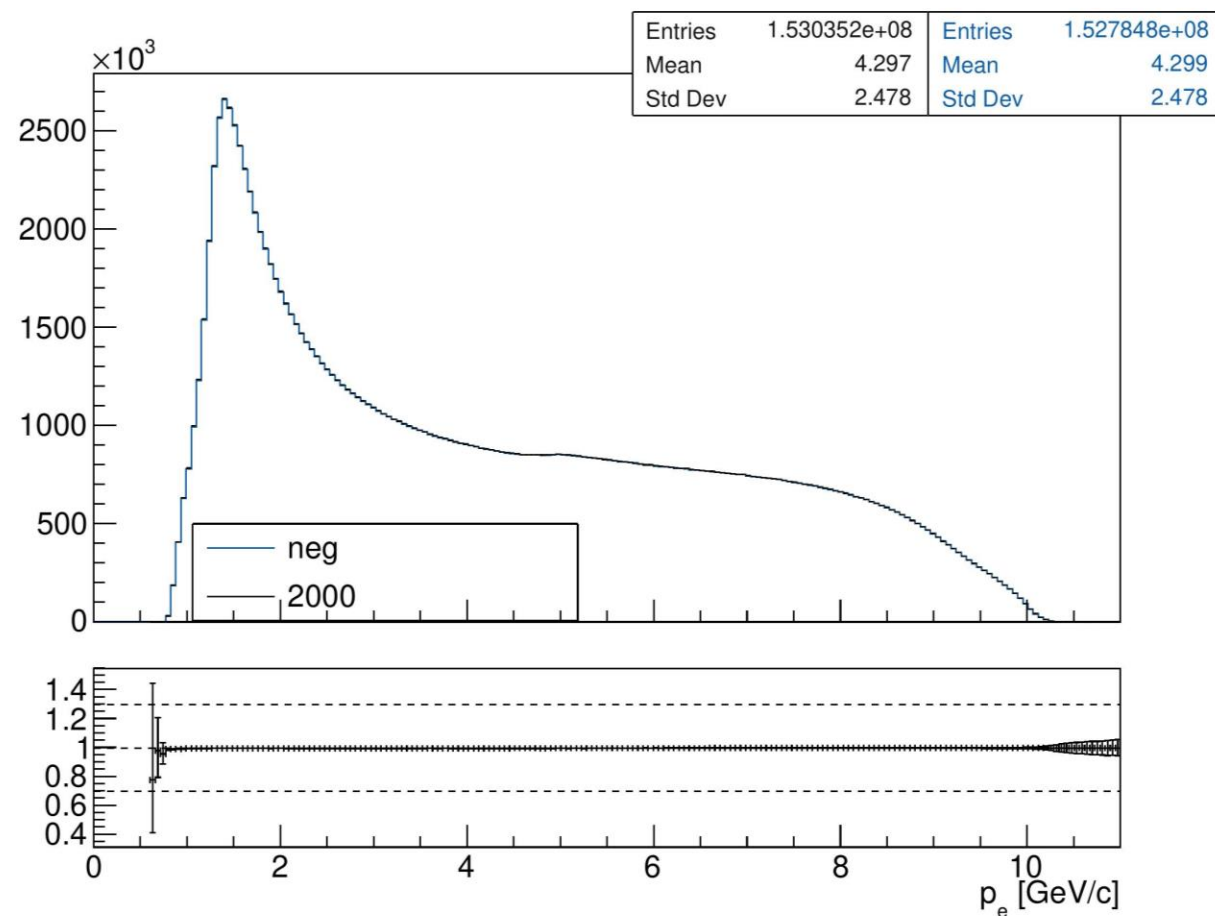
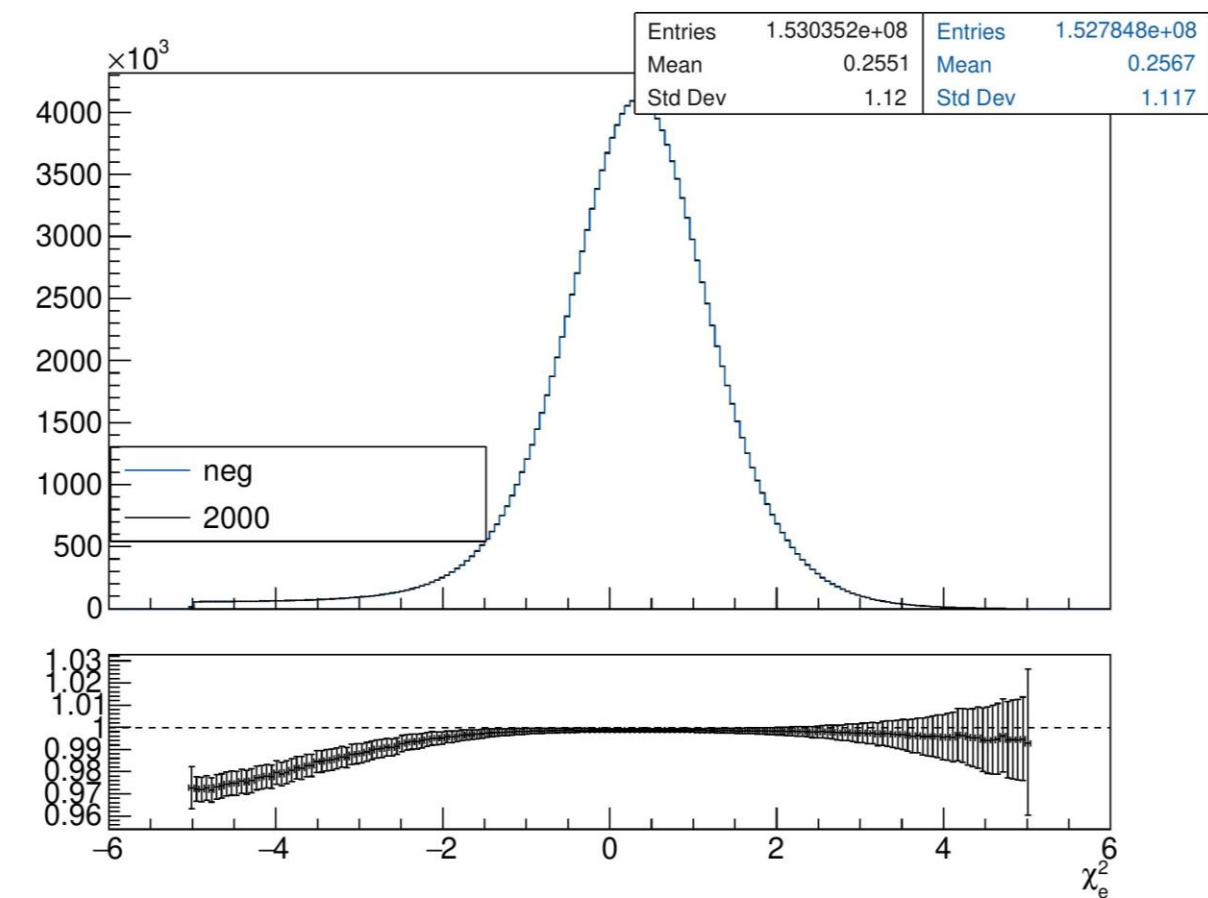
- Still have a higher dose for the FT Cal, around 4.5 rad/h
- To reduce that, we can remove the FT, reduce the luminosity (current: $2 \times 10^{34} \text{cm}^{-2}\text{s}^{-1}$), or run the FT at 20°C.



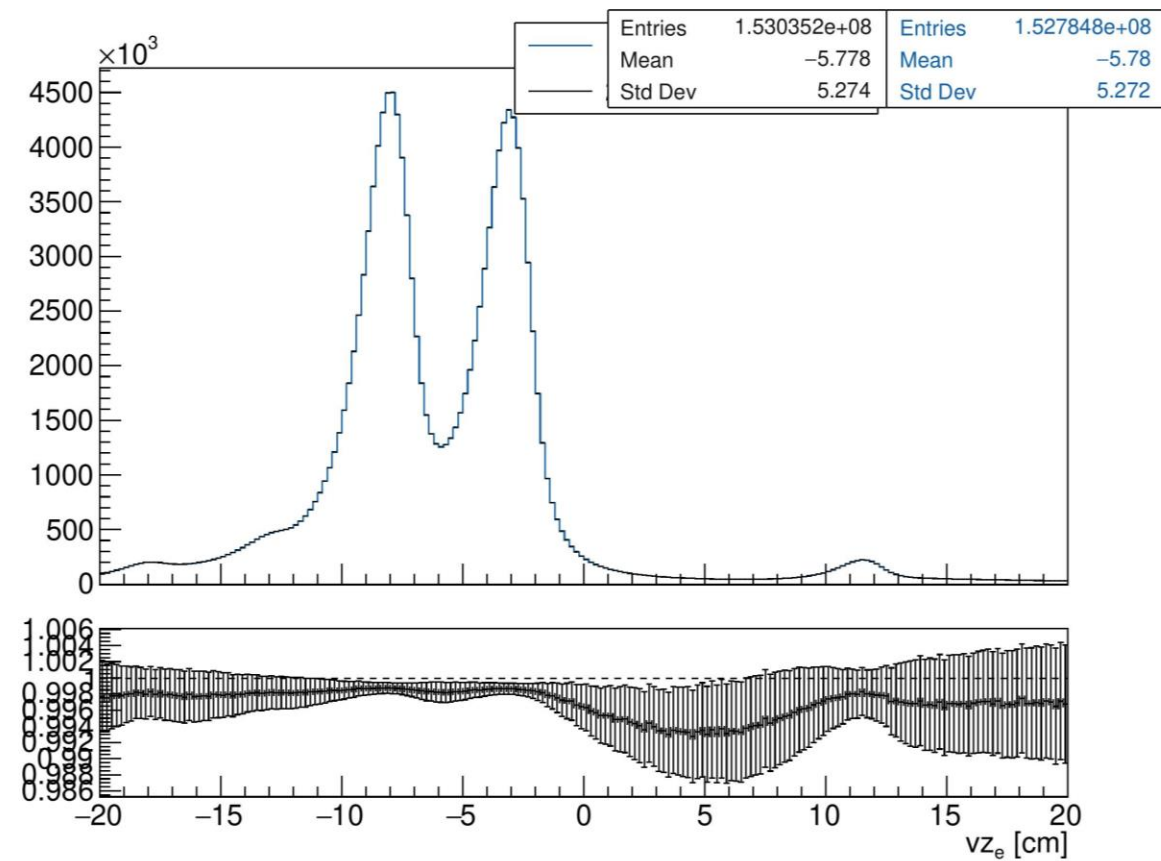
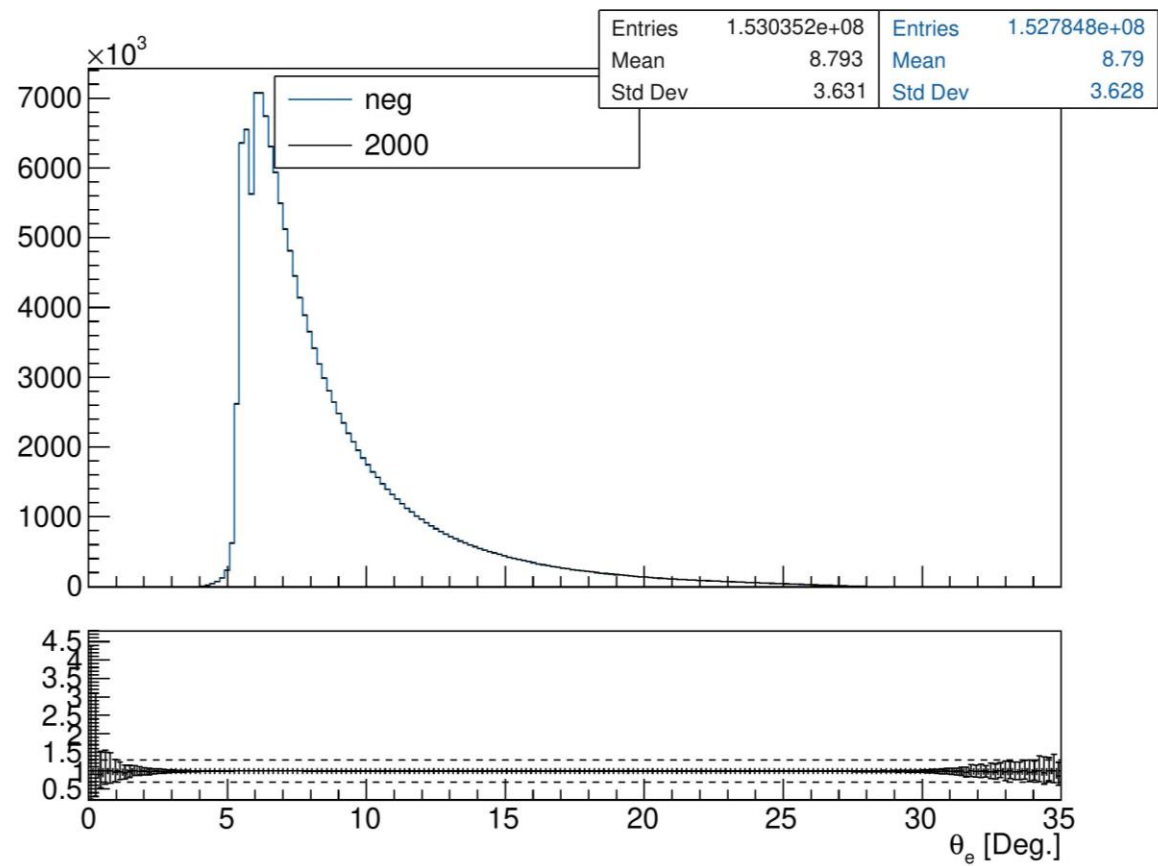
RG-D status study

- Goal : study the difference between selecting the electron using $\text{status} < 0$ or using $|\text{status}|/2000 = 1$ and take the most energetic.
- Use carbon runs 018451 and 018452.
- nb electron with a negative status : 152784791
- nb electron with a status modulo 2000 : 153035240
- Ratio : $152784791/153035240 = 0.998363455371$

RG-D status study



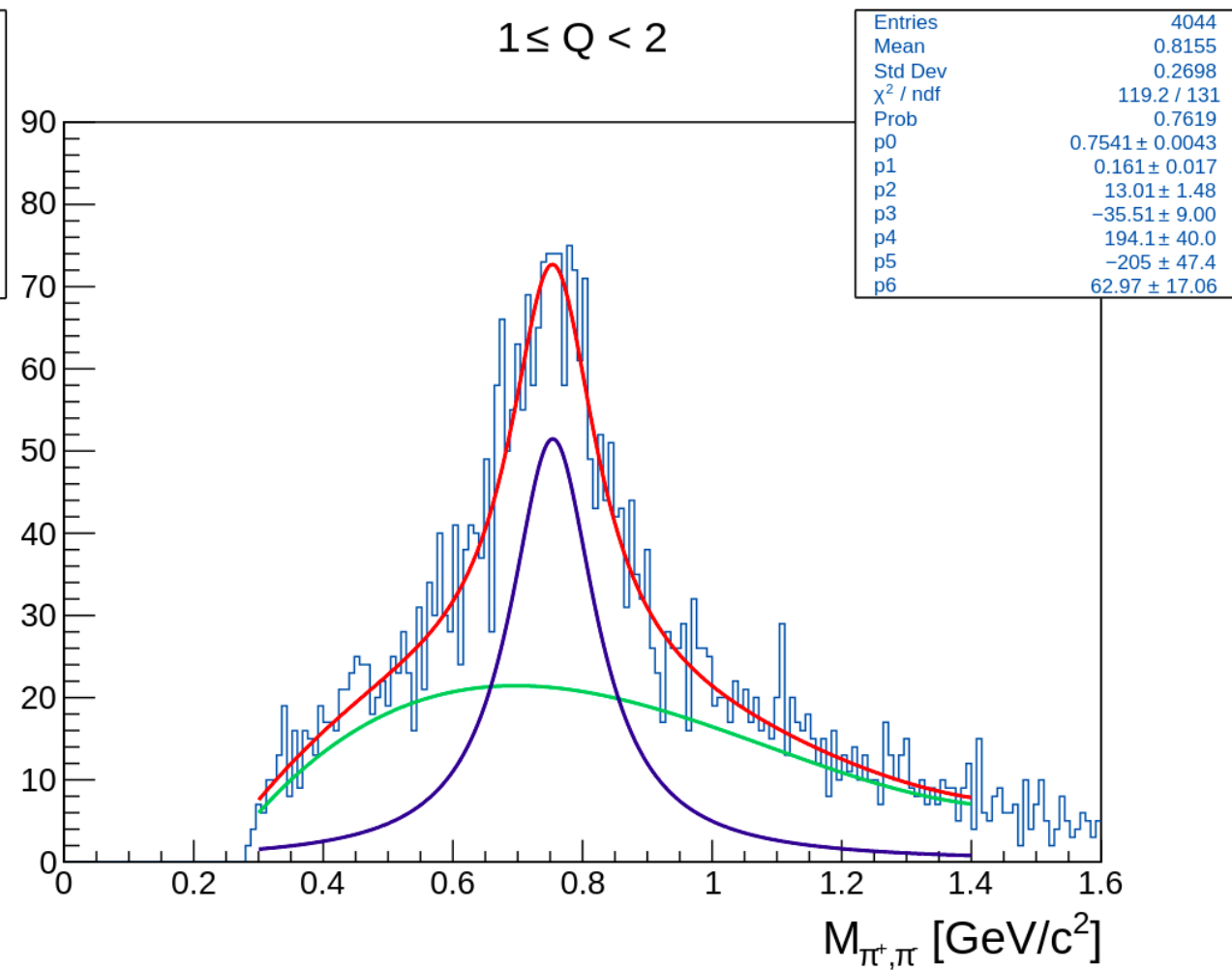
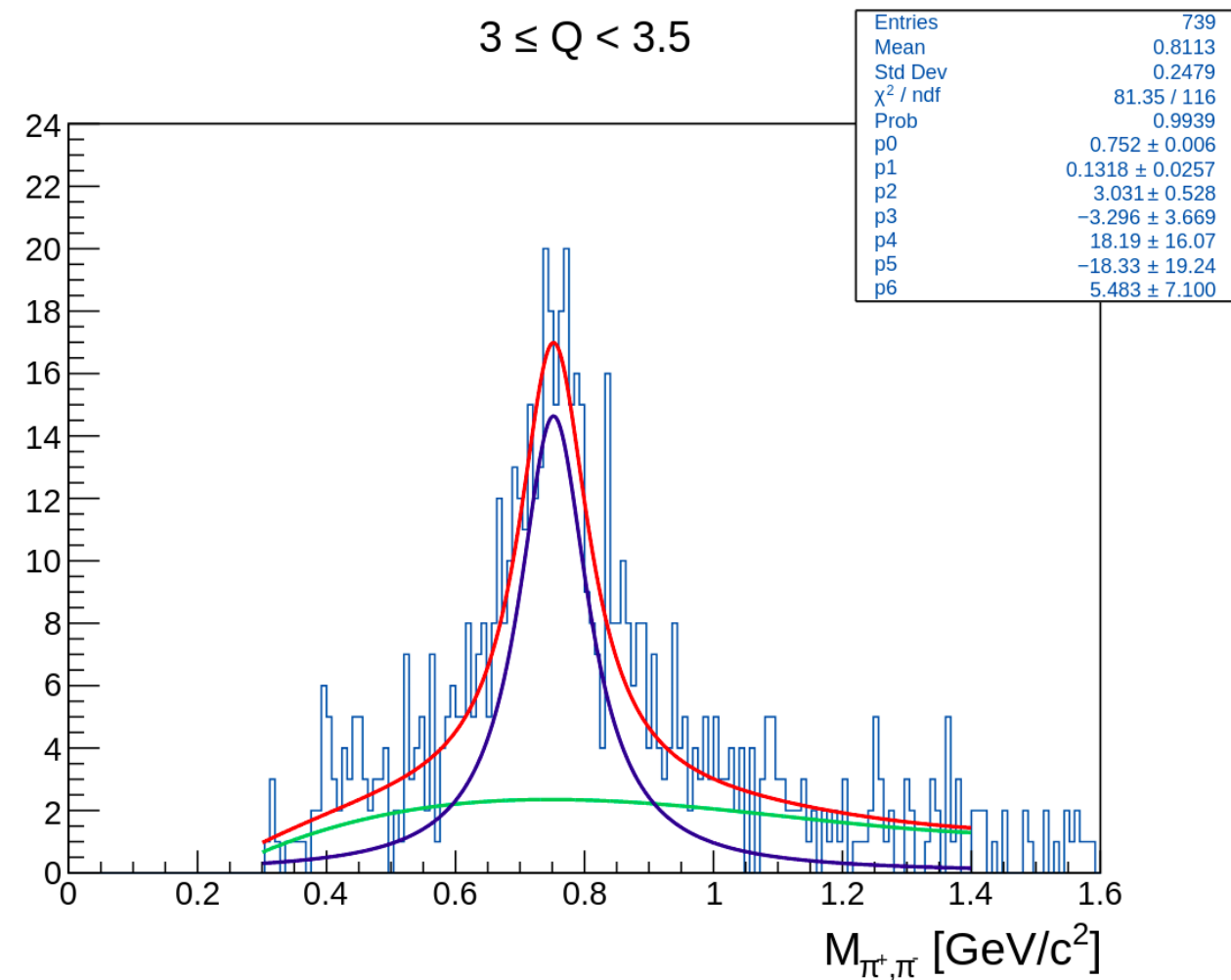
RG-D status study



Cross check Matt analysis

- Now we have the same number of electron, pion+ and pion-
- Same number of entries in the histograms of invariant mass of pion+ pion- for each bin in Q^2 .
- nb_pion_plus_forward : 9473911
- nb_pion_plus_central : 14205300
- nb_pion_minus_forward : 13755721
- nb_pion_minus_central : 8124497

Cross check Matt analysis



Cooking workflow for the timeline

- We can use AI for tracking and denoising for the timelines.
- The tag for the new timeline should be pass0v4 ?
- Which runs should I use (the last file available ?)